Math 2551 Worksheet: Arc Length

1. Let $\mathbf{r}(t) = \langle 6\sin 2t, 6\cos 2t, 5t\rangle$. Find the unit tangent vector of $\mathbf{r}(t)$ and find the length of the portion of the graph of $\mathbf{r}(t)$ where $0 \leq t \leq \pi$.

2. Find the point on the curve

$$\mathbf{r}(t) = (5\sin t)\mathbf{i} + (5\cos t)\mathbf{j} + 12t\mathbf{k}$$

   at a distance 26π units along the curve from the point $(0, 5, 0)$ in the direction of increasing arc length.

3. Suppose an object's position is given by $\mathbf{r}(t) = (2\ln(t+1))\mathbf{i} + (e^{2t} + t)\mathbf{j} + (\sin^2(t))\mathbf{k}$. Set up but do not evaluate the appropriate integral with limits to find the distance the object traveled from the point $A(0, 1, 0)$ to the point $B(\ln 4, e^2 + 1, \sin^2(1))$.

4. Find the length of the curve

$$\mathbf{r}(t) = \langle \sqrt{2}t, \sqrt{3}t, (1-t)\rangle$$

   from $(0, 0, 1)$ to $(\sqrt{2}, \sqrt{3}, 0)$.